ALAN WANG

E-mail: wangalan@berkeley.edu Website: https://urd00m.github.io github: https://github.com/urd00m

EDUCATION:

Ph.D. in Computer Science, University of California, Berkeley **2025-2030**

- **GPA:** 4.0 **Advisor:** Professor Christopher W. Fletcher

Conjoined BSMS in Computer Science, University of Illinois at Urbana-Champaign **2021-2025**

- **GPA:** 4.0 **Advisor:** Professor Christopher W. Fletcher

PUBLICATIONS:

- **Redacted**, under submission
- **Redacted,** first author, under submission
- **Pixnapping: Bringing Pixel Stealing out of the Stone Age**, first author, *ACM CCS '25,* https://www.pixnapping.com
- **EMT: An OS Framework for New Memory Translation Architectures**, *OSDI '25*
- **Peek-a-Walk: Leaking Secrets via Page Walk Side Channels**, first author, *IEEE S&P (Oakland) '25*
- **Declassiflow: A static analysis for modeling non-speculative knowledge to relax speculative execution security measures**, second author, *ACM CCS '23*

AWARDS/FELLOWSHIPS:

- **NDSEG Fellowship**, DoD, 2026
- **Apple Bug Bounty Award**, Apple, 2026
- **Micro Top Picks Honorable Mention,** IEEE, 2026
- **Jane Street GRF Finalist,** Jane Street, 2026
- **3x High-Severity Google Bug Bounty Awards**, Google, 2025
- **NSF GRFP Honorable Mention**, NSF, 2025
- **Bronze Tablet Scholar Award**, UIUC, 2025
- **Siebel Scholar Award**, 2025
- **Best Poster in Theme**, SRC PRISM, 2025
- **Dean's List**, UIUC's Grainger College, 2021-2023
- **2nd Place Overall**, NASA RASC-AL competition, 2022

RECENT EXPERIENCE (see CV for full list):

Research Assistant @ University of California, Berkeley **August 2025 – Current**

- Recent work: https://www.pixnapping.com. This works presents a new class of side-channel attacks on Android devices. Most notably, these attacks steal app-based 2FA codes in under 30 seconds
- Attacks: researching side-channel attacks inherent to graphics stacks (i.e., pixel stealing attacks)
- Defenses: exploring defenses for the side-channel vulnerabilities found by my attack research

Software Developer Intern @ Jane Street **May – August 2024**

- Improved the efficiency and utilization of server's used by the entire firm

Computer Science Intern @ D. E. Shaw Research **May – August 2023**

- Researched docking and non-equilibrium FE calculation methods
- Wrote system software which ran thousands of simulations on Anton3 ASICs and achieved a 2.5x performance improvement

Research Assistant @ FPSG Lab **September 2021 – May 2025**

- Created a new class of side-channel attacks on Android devices.
- Created a new general-purpose cache side channel exploiting page walkers and used it to exploit the Linux kernel using Spectre-V2 and the Intel DDP
- Modelled non-speculative information flow to decrease mitigation overhead by up to 21%

Research Aide @ Argonne National Lab **May 2022 – May 2023**

- Worked with NVIDIA's Bluefield-3 Data Processing Unit (DPU) for zero trust network architectures
- Found critical errors by instrumenting Portable Batch System (PBS) for Argonne's extreme scale systems
- Developed and programmed a command line interface for Argonne's UserBase3 and used by all Argonne admins

MENTORING/SERVICE:

- Berkeley Security Seminar Cohost (2026-current)
- Students mentoring: Jonathan Ji (2026-current), Steven Tang (2026-current), Zaheen Jamil (2026-current), Nathan Vaz (2026-current), Matin Sadeghian (2026-current), Ailsa Sun (2026-current), Kaustubh Khulbe (2025 spring), Ishan Buyyanapragada (2024-2025), Spencer Acquah (2023-2024), Christopher Suarez (2023-2024)
- ACM Mentor – Helping incoming freshman transition to college life, June 2022 – May 2025

BUGS/NEWS:

CVEs: CVE-2025-48561, CVE-2025-48630, CVE-2026-28964

Highlighted news: Ars Technica, Malwarebytes, Risky Business, and LWN